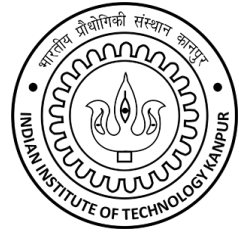




RESEARCH TRAINING REPORT



Design and Implementation of a Modulo Unit for the Posit Number System

Organization

CSIR–Central Electronics Engineering Research Institute (CEERI), Pilani

Training Mentor

Dr. Jai Gopal Pandey

Senior Principal Scientist

Societal Electronics Group (SEG)

Submitted by

Ishita Porwal

Trainee

B.Tech Student, IIT Kanpur

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **CSIR–CEERI, Pilani** for providing me with the opportunity to undergo winter training at their prestigious institute. The research-oriented environment and access to advanced facilities at CEERI played a crucial role in enhancing my practical understanding of digital hardware design and arithmetic system implementation.

I would like to convey my deepest appreciation to my training mentor, **Dr. Jai Gopal Pandey**, for his valuable guidance, constant support, and insightful suggestions throughout the training period. His technical expertise, encouragement, and patient mentoring were instrumental in helping me understand complex concepts related to the posit number system and its hardware realization. His guidance motivated me to approach problems with a systematic and research-driven mindset.

I would also like to extend my sincere thanks to **Ms. Diksha Shekhawat**, Ph.D. Scholar at CSIR–CEERI, for her continuous support and assistance during the course of this training. Her willingness to clarify concepts, provide practical insights, and offer constructive feedback greatly helped me in overcoming technical challenges encountered during the work.

Finally, I would like to express my gratitude to my institute and faculty members for encouraging me to undertake this training. The knowledge and experience gained during this training have significantly contributed to my academic growth and will be highly beneficial for my future studies and research work in the field of digital systems and VLSI design.

Contents

- Introduction
- Overview of Training Activities
- Learning Outcomes
- Conclusion

Introduction

This report presents the work carried out during my winter training at CSIR–CEERI, Pilani. The primary objective of this training was to gain practical exposure to digital hardware design methodologies and to explore advanced number systems employed in modern computing architectures. The training provided an opportunity to bridge the gap between theoretical knowledge and practical hardware implementation through hands-on design experience.

During the course of the training, emphasis was placed on understanding the limitations of conventional number representations and studying alternative number systems that offer improved numerical efficiency. In this context, the Posit number system was explored in detail due to its advantages in terms of dynamic range and accuracy.

The main focus of the training was the design and implementation of a modulo (MOD) arithmetic unit for posit numbers, along with the study and implementation of basic supporting arithmetic blocks. All designs were developed using Verilog HDL at the register-transfer level (RTL), with an emphasis on sequential, FSM-based architectures suitable for hardware realization. Functional correctness and design behavior were verified through simulation

Overview of Training Activities

Study of Posit Number System

During the initial phase of the training, the posit number system was studied with emphasis on its fundamental components, including the sign bit, variable-length regime field, exponent, and mantissa. Particular attention was given to the role of the regime field in enabling a wide dynamic range through run-length encoding.

The encoding and decoding of posit numbers were analyzed to understand their hardware interpretation. A comparison with the IEEE-754 floating-point representation was also carried out, highlighting the advantages of posits in terms of improved accuracy and dynamic range, thereby providing a theoretical foundation for subsequent hardware design activities.

Design of Sequential Adder

As part of the training, a sequential adder architecture was studied and implemented using a finite state machine (FSM) based approach. The design was developed to perform addition in a step-by-step manner, allowing a clear understanding of control flow, data path interaction, and timing behavior in sequential digital systems. The implementation was carried out using basic digital logic concepts, and the functionality of the adder was verified through simulation and experimental testing. The design was further implemented and tested on a Boolean Algebra Board, which provided practical exposure to hardware realization, signal observation, and verification of sequential operations under real hardware constraints.

Design of Wallace Tree Multiplier

A Wallace tree multiplier was designed and studied to understand efficient hardware implementation of multiplication operations. The Wallace multiplier architecture reduces the partial products generated during multiplication using a tree of carry-save adders, thereby minimizing the number of sequential addition stages required. This approach significantly improves computational speed compared to conventional array multipliers by performing partial product reduction in parallel.

Design of Sequential Posit Modulo Unit

The primary focus of the training was the design and implementation of a sequential modulo (MOD) unit for posit numbers. The modulo operation was implemented using a shift-and-subtract algorithm controlled by a finite state machine.//

1. Field Extraction (*data_extract*)

Before any arithmetic operation can be performed, the Posit number must be decoded into its constituent fields.

Regime Detection: A Hierarchical Leading One Detector (LOD) is used to determine the length of the regime field. The detection is performed in stages:

$$\text{LOD8} \rightarrow \text{LOD16} \rightarrow \text{LOD32} \rightarrow \text{LOD64}$$

Field Parsing: Once the regime length is identified, a Barrel Shifter is used to left-shift the Posit word. This alignment allows the Exponent and Mantissa fields to be extracted correctly.

2. Core Modulus Logic (*mod_fsm*)

The modulus operation is carried out on the extracted mantissas using a Restoring Division algorithm implemented as a Finite State Machine (FSM).

Initial Comparison: If the magnitude of operand A is smaller than that of B , the modulus result is simply A , and the FSM bypasses further computation.

Iterative Subtraction: The FSM repeatedly subtracts the divisor from the remainder and performs a left shift. This process is repeated for a number of cycles equal to the exponent difference.

Normalization: After the iterative process, the remainder is normalized by shifting it until it reaches the standard mantissa format. The exponent is adjusted accordingly to reflect this normalization.

3. Reconstruction (*posit_mod*)

The final stage converts the normalized result back into a valid Posit representation.

Descaling: The resulting mantissa and scale obtained from the FSM are decomposed into the Posit-specific Regime value k and Exponent value e .

Bit Packing: The final Posit bitstream is constructed by concatenating the fields in the following order:

$$\text{Sign} \mid \text{Regime} \mid \text{Exponent} \mid \text{Mantissa}$$

Sign Handling: If the original operand was negative, a two's complement operation is applied to the final Posit result to preserve the correct sign.

Learning Outcomes

The training helped in developing the following learning outcomes:

- A clear understanding of the posit number system, its internal representation, and the role of sign, regime, exponent, and mantissa in arithmetic operations.
- Insight into the hardware implications of posit arithmetic, particularly the challenges associated with variable-length fields, normalization, and re-encoding.
- Practical experience in designing and implementing sequential arithmetic units using finite state machine (FSM) based control and datapath separation.
- Hands-on exposure to the design of a sequential modulo (MOD) operation for posit numbers using shift-and-subtract techniques and fixed-point mantissa interpretation.
- Improved proficiency in Verilog HDL, including RTL coding, parameterized design, and simulation of sequential digital systems.
- Understanding of normalization techniques such as leading bit detection and scaling adjustments required in arithmetic hardware.
- Exposure to the fundamentals of Post-Quantum Cryptography (PQC) and the importance of efficient modular arithmetic in cryptographic standards such as FIPS 203.
- Enhanced problem-solving skills and the ability to translate mathematical algorithms into hardware-oriented implementations.

Conclusion

The winter training at CSIR–CEERI, Pilani was a valuable and enriching learning experience that provided practical exposure to advanced concepts in digital hardware design. The training offered an opportunity to apply theoretical knowledge to real-world hardware implementation challenges in a research-oriented environment.

During the training, significant emphasis was placed on understanding the posit number system and its internal structure, including regime-based scaling and normalization. The design and implementation of a sequential modulo (MOD) unit for posit numbers using FSM-based control helped in developing a clear understanding of iterative arithmetic operations, normalization techniques, and hardware-efficient RTL design using Verilog HDL. The study and implementation of basic arithmetic blocks such as adder and multiplier further strengthened the foundation of sequential logic and data-path control.

In addition to arithmetic design, exposure to Post-Quantum Cryptography (PQC) and the NIST standard FIPS 203 provided valuable insight into the importance of efficient modular arithmetic in cryptographic hardware systems. This broadened the scope of learning by connecting arithmetic unit design with emerging security applications.

Overall, the training enhanced my technical skills in Verilog-based RTL design, finite state machine architectures, and arithmetic hardware implementation. It also strengthened my interest in digital systems, VLSI design, and research in advanced number systems. The knowledge and experience gained during this training will serve as a strong foundation for future academic work and professional pursuits in hardware design and related research areas.